Env ⇒ quantum_control

Repo ⇒ Quantum_Resonator_Control

Lorentzian Response Function ⇒ $A(f) = \dfrac{A_{max}}{1 + Q^2\left(\frac{f - f_0}{f_0}\right)^2}$

→ used to describe Resonance

(Cauchy's Distribution)

⇒ Actual signal response & Reward system

A_{max} → Max Amplitude (Peak height)

Q → Quality factor (used to measure the sharpness of the curve)

f → frequency

f_0 → Resonant frequency

In order to maximize energy (for high SNR (Signal to Noise Ratio)) transfer into the system, f must be equal to f_0 such that $(f - f_0) = 0$ which helps us to get the max O/P $A(f) = A_{max}$

f_0 here is a hidden value & the AI doesn't have access to the value of f_0.

f_0 is unstable due to $\underbrace{\text{temp., power supply jitters \& component aging}}_{\text{Noise}}$

AI → Reinforcement Layer ⇒ Proximal Policy Optimization (PPO) algorithm

→ autonomous calibration layer → based off of Amplitude response

Reward

Action	+ve	-ve
+ve	Move Right	Move Left
-ve	Move Left	Move Right

cont. Switch

$\Delta f \Rightarrow$ Change in frequency

If $A(f+\Delta f) > A(f)$; closer to the peak → present at the left side → f should increase to reach peak $\left(\frac{dA}{df} > 0\right)$

If $A(f+\Delta f) < A(f)$; farther from peak → present at the right side → f should decrease to reach peak $(dA/df < 0)$

→ Robust & Hence good for envs with noise

Proximal Policy Optimization ⇒ RL algorithm

- → Policy → Controller ⇒ I/P = Amp O/P = Δf → Learns the best action to take to reach the resonance peak
- → Value → Observer ⇒ Predicts the reward

Resonant Frequency ⇒ $f_0 = \dfrac{1}{2\pi\sqrt{LC}}$